

MP491 2020/2021: Tutorial Sheet 2

2-dim linear ODEs

I. Explicit solutions

For each of the following systems of ODEs identify the nature and stability of the equilibrium point at $(0, 0)$ by analysing the coefficient matrix A . Find the general solutions $x(t)$ and $y(t)$ in each case.

1.

$$\begin{aligned}\frac{dx}{dt} &= 2x - 4y, \\ \frac{dy}{dt} &= 2x - 2y,\end{aligned}$$

2.

$$\begin{aligned}\frac{dx}{dt} &= 3x + y, \\ \frac{dy}{dt} &= -x + 5y,\end{aligned}$$

3.

$$\begin{aligned}\frac{dx}{dt} &= 4x + 2y, \\ \frac{dy}{dt} &= 3x - y,\end{aligned}$$

II. Phase plane portraits

For each of the following systems of ODEs identify the nature and stability of the equilibrium point at $(0, 0)$ by analysing the coefficient matrix A . Deduce the xy -phase plane portrait in each case.

1.

$$\begin{aligned}\frac{dx}{dt} &= 4x - 2y, \\ \frac{dy}{dt} &= 3x - y,\end{aligned}$$

2.

$$\begin{aligned}\frac{dx}{dt} &= x - 3y, \\ \frac{dy}{dt} &= 4x - 2y,\end{aligned}$$

3.

$$\begin{aligned}\frac{dx}{dt} &= 3x - y, \\ \frac{dy}{dt} &= x + y,\end{aligned}$$

III. Parameter dependence

Each of the following systems of ODEs involves a real parameter a . Investigate how the nature of the equilibrium point at $(0, 0)$ changes as this parameter varies along the real line.

1.

$$\begin{aligned}\frac{dx}{dt} &= 3x + ay, \\ \frac{dy}{dt} &= x + y,\end{aligned}$$

2.

$$\begin{aligned}\frac{dx}{dt} &= ax - y, \\ \frac{dy}{dt} &= 2x - 4y,\end{aligned}$$

3.

$$\begin{aligned}\frac{dx}{dt} &= -4x - 2y, \\ \frac{dy}{dt} &= ax + y,\end{aligned}$$